

Student Number: \_\_\_\_\_

**St George Girls High School**  
**Trial Higher School Certificate Examination**  
**2019**



# Mathematics Extension 2

## General Instructions

- Reading time – 5 minutes
- Working time – 3 hours
- Write using black pen
- Board-approved calculators may be used
- A reference sheet is provided
- In Questions 11 – 16, show relevant mathematical reasoning and/or calculations

|                   |             |
|-------------------|-------------|
| <b>Section I</b>  | <b>/10</b>  |
| <b>Section II</b> |             |
| Question 11       | /15         |
| Question 12       | /15         |
| Question 13       | /15         |
| Question 14       | /15         |
| Question 15       | /15         |
| Question 16       | /15         |
| <b>Total</b>      | <b>/100</b> |

## Total Marks – 100

### Section I

Pages 2 – 5

#### 10 marks

- Attempt Questions 1 – 10
- Allow about 15 minutes for this section
- Answer on the multiple choice answer sheet provided at the back of this paper

### Section II

Pages 6 – 11

#### 90 marks

- Attempt Questions 11 – 16
- Allow about 2 hours and 45 minutes for this section
- Begin each question in a new writing booklet

## Section I

10 marks

Attempt Questions 1 – 10

Allow about 15 minutes for this section

Use the multiple-choice answer sheet for Questions 1–10.

1. If  $z = \frac{3+4i}{1+2i}$ , the imaginary part of  $z$  is:

(A)  $-2$

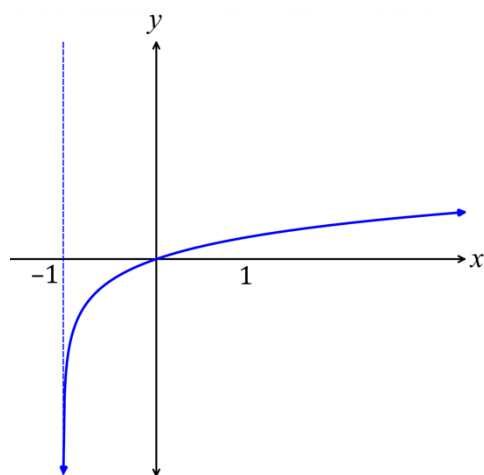
(B)  $-\frac{2}{5}$

(C)  $-\frac{2}{5}i$

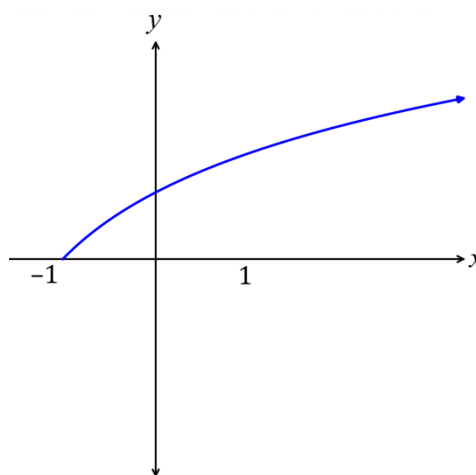
(D)  $-2i$

2. Which graph best represents  $y = |\log_e(x + 1)|$ ?

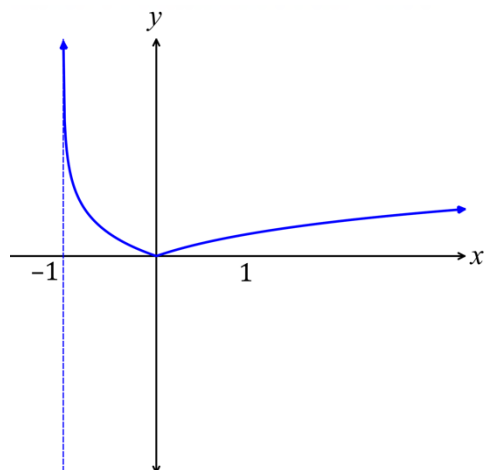
(A)



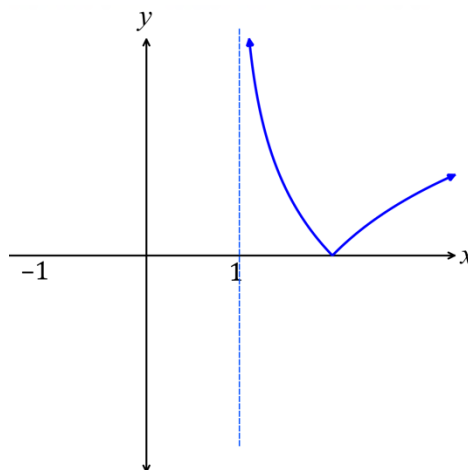
(B)



(C)



(D)



**Section I (cont'd)**

3. The equation  $2x^3 - 4x^2 - 8x - 1 = 0$  has roots  $\alpha$ ,  $\beta$  and  $\gamma$ .

What is the value of  $\alpha^{-3}\beta^{-3}\gamma^{-3}$ ?

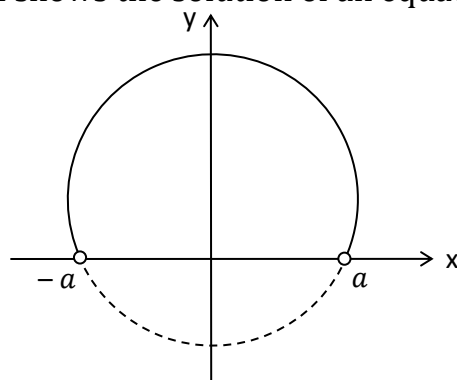
(A)  $-8$

(B)  $-\frac{1}{8}$

(C)  $\frac{1}{8}$

(D)  $8$

4. The diagram shows the solution of an equation.



Which of these could be the equation, if  $z$  is a point on the circumference of the circle?

(A)  $\text{Arg}(z - a) = \text{Arg}(z + a)$

(B)  $\text{Arg}(z - a) - \text{Arg}(z + a) = \frac{\pi}{2}$

(C)  $\text{Arg}(z + a) - \text{Arg}(z - a) = \frac{\pi}{3}$

(D)  $\text{Arg}(z - a) - \text{Arg}(z + a) = \frac{\pi}{3}$

**Section I (cont'd)**

5. Find  $\int \frac{dx}{x^2 + 4x + 9}$ .

(A)  $\frac{1}{\sqrt{5}} \tan^{-1} \left( \frac{x+2}{\sqrt{5}} \right) + c$

(B)  $\frac{1}{5} \tan^{-1} \left( \frac{x+2}{\sqrt{5}} \right) + c$

(C)  $\frac{1}{\sqrt{5}} \tan^{-1} \left( \frac{x}{\sqrt{5}} \right) + c$

(D)  $\frac{1}{5} \tan^{-1} \left( \frac{x}{\sqrt{5}} \right) + c$

6. The polynomial  $P(x) = x^3 - 6x^2 + 9x + c$  has a double zero.  
What are the values of  $c$ ?

(A)  $c = -4$  or  $c = -1$

(B)  $c = -4$  or  $c = 0$

(C)  $c = 4$  or  $c = 7$

(D)  $c = 4$  or  $c = 0$

7. Which of the following is an expression for  $\int xe^{-x} dx$ ?

(A)  $-xe^{-x} - e^{-x} + c$

(B)  $-xe^{-x} + e^{-x} + c$

(C)  $xe^{-x} - e^{-x} + c$

(D)  $xe^{-x} + e^{-x} + c$

**Section I (cont'd)**

8. What is the eccentricity of the ellipse  $4x^2 + 9y^2 = 16$ ?

(A)  $\frac{13}{3}$

(B)  $\frac{\sqrt{5}}{9}$

(C)  $\frac{\sqrt{13}}{3}$

(D)  $\frac{\sqrt{5}}{3}$

9. The circle  $x^2 + y^2 = 9$  is rotated about the line  $x = 4$  to form a solid.  
What is an expression for the volume of the solid using the method of cylindrical shells?

(A)  $2\pi \int_0^3 (4 - x)\sqrt{9 - x^2} dx$

(B)  $4\pi \int_0^3 (4 - x)\sqrt{9 - x^2} dx$

(C)  $2\pi \int_{-3}^3 (4 - x)\sqrt{9 - x^2} dx$

(D)  $4\pi \int_{-3}^3 (4 - x)\sqrt{9 - x^2} dx$

10. A body of mass  $m$  kg moves in a straight line with initial speed  $U$  ms<sup>-1</sup> subject to a resistance of magnitude  $m(1 + v)$  Newtons when its speed is  $v$  ms<sup>-1</sup>. What is the time taken in seconds by the body to come to rest?

(A)  $\frac{1}{1 + U}$

(B)  $\sqrt{1 + U}$

(C)  $\ln(1 + U)$

(D)  $e^{1+U}$

**End of Section I**

**Section II**

**90 marks**

**Attempt Questions 11 – 16**

**Allow about 2 hours 45 minutes for this section**

Answer each question in a SEPARATE writing booklet. Extra writing booklets are available.

In Questions 11 – 16, your responses should include relevant mathematical reasoning and/or calculations.

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| Question 11 (15 marks) Use a SEPARATE writing booklet                                                | Marks |
|------------------------------------------------------------------------------------------------------|-------|
| a) For $z = 3 + i$ and $w = 2 + 2i$ , find the values of:                                            |       |
| (i) $\arg(w - z)$                                                                                    | 1     |
| (ii) $w + \overline{2iz}$                                                                            | 2     |
| b) Use a suitable trigonometric substitution to evaluate the following integral.                     | 3     |
| $\int_0^1 \sqrt{1 - x^2} dx$                                                                         |       |
| c) (i) Express $\frac{x}{(x + 1)(x^2 + 1)}$ in the form $\frac{A}{x + 1} + \frac{Bx + C}{x^2 + 1}$ . | 2     |
| (ii) Hence find $\int \frac{x}{(x + 1)(x^2 + 1)} dx$ .                                               | 2     |
| d) (i) Find real numbers $a$ and $b$ such that $(a + ib)^2 = -3 + 4i$ .                              | 3     |
| (ii) Hence solve the equation $z^2 - 3z + (3 - i) = 0$ .                                             | 2     |

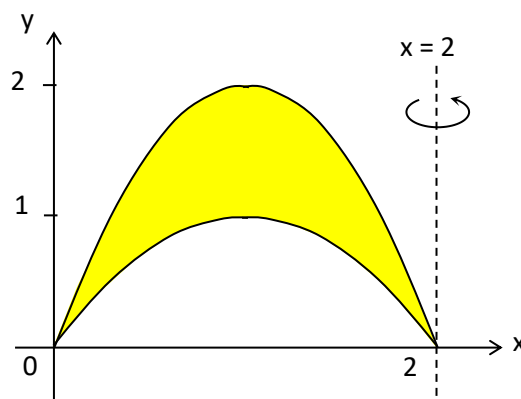
**Question 12** (15 marks)      Use a SEPARATE writing booklet **Marks**

- a) A rectangular hyperbola  $A$  has the equation  $xy = 18$ .
- (i) Show that the equation of the normal  $N_1$  to  $A$  at the point  $P(2, 9)$  is given by  $2x - 9y + 77 = 0$ . 2
- (ii) Find, in general form, an equation of the normal  $N_2$  to  $A$  at the point  $Q(9, 2)$ . 1
- (iii) Find the coordinates of point  $B$  where  $N_1$  and  $N_2$  intersect. 1
- (iv) Show that the quadrilateral  $OPBQ$  is a rhombus. 2
- b) Find all the roots of the equation: 3

$$18x^3 + 3x^2 - 28x + 12 = 0$$

given that two of the roots are equal.

- c) The shaded region bounded by the parabolas  $y = 2x - x^2$  and  $y = 4x - 2x^2$  between  $x = 0$  and  $x = 2$  is as shown in the diagram. 3



Use the cylindrical shells method to find the volume of the solid formed when the shaded area is rotated about the line  $x = 2$ .

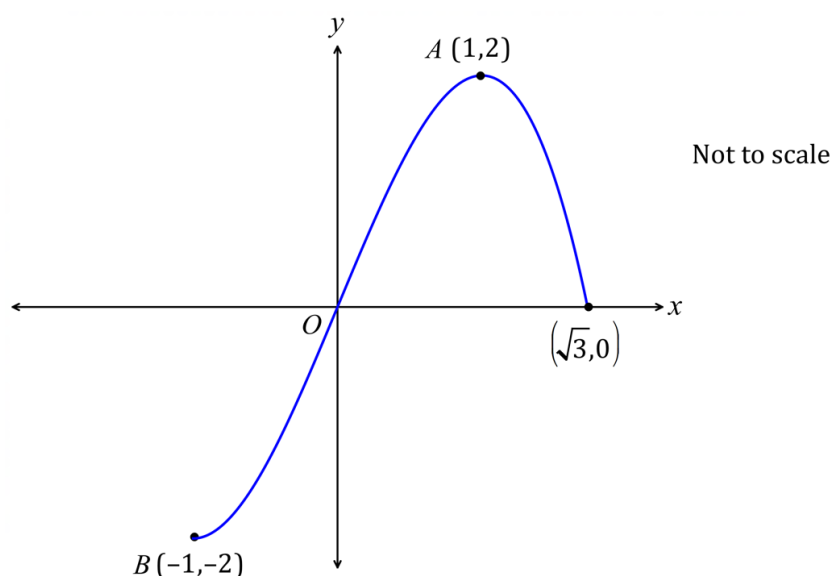
- d) On an Argand diagrams, shade in the regions containing all points representing complex numbers  $z$  such that: 3

$$|z| \leq |z - 2| \text{ and } -\frac{\pi}{4} \leq \arg z \leq \frac{\pi}{4}$$

**Question 13** (15 marks) Use a SEPARATE writing booklet.

**Marks**

a)



The graph of  $y = f(x)$  is shown above. Draw separate one-third page sketches of these functions. Indicate clearly any asymptotes and intercepts with the axes.

(i)  $y = f(|x|)$  2

(ii)  $y = \sqrt{f(x)}$  2

(iii)  $y = [f(x)]^2$  2

(iv)  $y = f^{-1}(x)$  2

- b) A solid is formed by rotating about the  $x$  –axis, the region bounded by the parabola  $y^2 = 4ax$ , the  $x$  –axis and ordinate  $x = a$ . Find the volume of this solid using the method of slicing. 3

c)  $I_n = \int_1^e (1 - \log_e x)^n dx \quad n = 0, 1, 2, \dots$

(i) Show that  $I_n = -1 + nI_{n-1} \quad n = 1, 2, 3, \dots$  2

(ii) Hence evaluate  $\int_1^e (1 - \log_e x)^3 dx$ . 2

**Question 14** (15 marks) Use a SEPARATE writing booklet.

**Marks**

- a) A particle of mass  $m$  falls from rest in a medium where the resistance to motion has magnitude  $mkv$  for some positive constant  $k$  when the speed is  $v$ , and the terminal velocity of the particle is  $V$ . The particle falls a distance  $x$  in time  $t$ , and the acceleration due to gravity is  $g$ .

(i) Show that  $\ddot{x} = \frac{g}{V}(V - v)$ . 2p

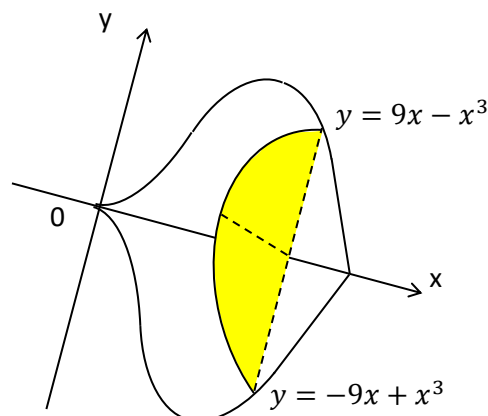
(ii) If the particle attains half its terminal velocity in time  $T$  and falls a distance  $X$  in this time, show that  $VT - X = \frac{V^2}{2g}$ . 4

b) Sketch the graph of  $y = x^2 + \frac{1}{x^2}$ . 3

c) Find the equation of the tangent to the curve  $x^2y + xy^2 - 6 = 0$  at the point  $(1, 2)$ . 3

d) The base of a solid is the region bounded by the curves  $y = 9x - x^3$ ,  $y = -9x + x^3$  between  $x = 0$  and  $x = 3$ . 3

Each cross section perpendicular to the  $x$ –axis is a semi circle as shown in the diagram.



Find the volume of this solid.

**Question 15** (15 marks) Use a SEPARATE writing booklet.

**Marks**

- a)  $P(a\cos\theta, b\sin\theta)$  lies on the ellipse  $\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$   $a > b > 0$ .

The tangent and the normal at  $P$  cut the  $y$  – axis at  $A$  and  $B$  respectively, and  $S(ae, 0)$  is a focus of the ellipse.

- (i) Show that the tangent to the ellipse at  $P$  is the following equation. 2

$$\frac{x\cos\theta}{a} + \frac{y\sin\theta}{b} = 1$$

- (ii) Show that the normal to the ellipse at  $P$  is the following equation. 2

$$\frac{ax}{\cos\theta} - \frac{by}{\sin\theta} = a^2 - b^2$$

- (iii) Why is  $\angle ASB = 90^\circ$ ? 3

- (iv) Deduce that  $A, P, S$  and  $B$  are concyclic points. 1

- b) A particle is initially at  $x = 1$  with a velocity of 2 m/s.

The acceleration of the particle is given by  $a = \frac{1}{2} \left( 1 - \frac{1}{x^2} \right)$ , where  $x$  is the displacement of the particle from 0.

- (i) Prove that  $\frac{dx}{dt} = \frac{1+x}{\sqrt{x}}$ . 3

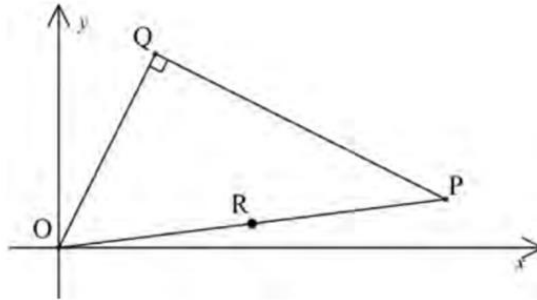
- (ii) Find the time taken for the particle to reach  $x = 3$ . 4

**Question 16** (15 marks) Use a SEPARATE writing booklet.

**Marks**

- a) In the Argand diagram below,  $OPQ$  is a triangle, which is right-angled at  $Q$ .

The point  $R$  is the midpoint of  $OP$ .



- (i) If  $\overrightarrow{OP} = z$  and  $\overrightarrow{OQ} = w$  show that  $\overrightarrow{OR} = \frac{1}{2}(1 - ki)w$ ,  
 where  $k$  is a constant,  $k > 0$ .

3

- (ii) Express  $\overrightarrow{RQ}$  in terms of  $w$  and hence show  $|\overrightarrow{OR}| = |\overrightarrow{RQ}|$ .

2

- b) If  $x > 1$ , show that  $\tan^{-1}\left(\frac{x+1}{x-1}\right) - \tan^{-1}\left(\frac{1}{x}\right) = \frac{\pi}{4}$ .

3

- c) A particle of mass  $m$  moves along the  $x$ -axis, beginning at  $x = 0$ .

It experiences a resistive force  $R$  given by  $R = kv$ , where  $k$  is a constant and  $v$  is the velocity of the particle.

- (i) Show that its speed  $v$ , is given by  $v = v_0 e^{-\frac{kt}{m}}$ , where  $v_0$  is the initial speed.

3

- (ii) Show that the displacement ( $x$ ) of the particle after  $t$  seconds is given by

3

$$x = \frac{mv_0}{k} \left[ 1 - e^{-\frac{kt}{m}} \right].$$

- (iii) Show that its limiting position  $x_L$  is given by  $x_L = \frac{mv_0}{k}$ .

1

## MATHEMATICS EXTENSION 2 - QUESTION MC

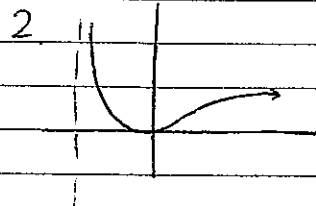
SUGGESTED SOLUTIONS

MARKS

MARKER'S COMMENTS

$$\begin{aligned}
 1. \quad z &= \frac{3+4i}{1+2i} \cdot \frac{1-2i}{1-2i} \\
 &= \frac{3+4i-6i+8}{5} \\
 &= \frac{11-2i}{5} \\
 &= \frac{11}{5} - \frac{2}{5}i
 \end{aligned}$$

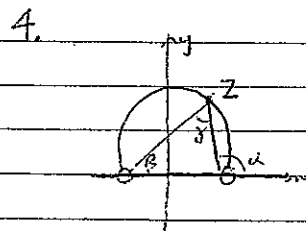
B



C

$$\begin{aligned}
 3. \quad \alpha \beta \gamma &= \frac{1}{2} \\
 \alpha^{-3} \beta^{-3} \gamma^{-3} &= (\alpha \beta \gamma)^{-3} \\
 &= \left(\frac{1}{2}\right)^{-3} \\
 &= 8
 \end{aligned}$$

D



$\arg(z-a) = \alpha$  and  $\arg(z+a) = \beta$   
 then  $\alpha = \beta + \gamma$  (ext  $\angle$ 's property)

$$\text{i.e. } \arg(z-a) = \arg(z+a) + \gamma$$

$$\text{So } \arg(z-a) - \arg(z+a) = \gamma$$

But  $\angle$  at centre is less than  $\pi$   
 $\therefore \angle$  at  $z$  must be less than  $\frac{\pi}{2}$   
 $\therefore \arg(z-a) - \arg(z+a) = \frac{\pi}{2}$

D

## MATHEMATICS EXTENSION 2 - QUESTION MC

MC

SUGGESTED SOLUTIONS

MARKS

MARKER'S COMMENTS

$$\begin{aligned}
 5. \quad \int \frac{dx}{x^2+4x+9} &= \int \frac{dx}{x^2+4x+4+5} \\
 &= \int \frac{dx}{(x+2)^2+5} \\
 &= \frac{1}{\sqrt{5}} \tan^{-1}\left(\frac{x+2}{\sqrt{5}}\right) + C
 \end{aligned}$$

A

$$\begin{aligned}
 6. \quad P(x) &= x^3 - 6x^2 + 9x + c \\
 P'(x) &= 3x^2 - 12x + 9 \\
 &= 3(x^2 - 4x + 3) \\
 &= 3(x-3)(x-1)
 \end{aligned}$$

$P'(x) = 0$  and  $P(x) = 0$  for a double root.

$$\begin{aligned}
 \text{For } x=3, \quad P(3) &= 0 \\
 P(3) &= 3^3 - 6(3)^2 + 9(3) + c \\
 0 &= 27 - 54 + 27 + c \\
 c &= 0
 \end{aligned}$$

$$\begin{aligned}
 P(1) &= 0 \\
 0 &= 1 - 6(1) + 9(1) + c \\
 c &= -4
 \end{aligned}$$

B

$$7. \quad \int x e^{-x} dx$$

Integrating by Parts

$$\begin{aligned}
 I &= -x \cdot e^{-x} - \int -e^{-x} dx \\
 &= -x e^{-x} + \int e^{-x} dx \\
 &= -x e^{-x} - e^{-x} + C
 \end{aligned}$$

A

# MATHEMATICS EXTENSION 2 – QUESTION MC

SUGGESTED SOLUTIONS

MARKS

MARKER'S COMMENTS

$$8. \quad 4x^2 + 9y^2 = 16$$

$$\frac{x^2}{4} + \frac{y^2}{16} = 1$$

$$a^2 = 4 \quad b^2 = 16/9$$

$$b^2 = a^2 (1 - e^2)$$

$$\frac{16}{9} = 4(1 - e^2)$$

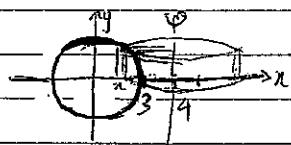
$$\frac{16}{36} = 1 - e^2$$

$$e^2 = \frac{5}{9}$$

$$e = \frac{\sqrt{5}}{3}$$

D

9. By Cylindrical Shells



Outer radius =  $4 - x + \delta x$

Inner radius =  $4 - x$

$$\begin{aligned} \delta V &= 2\pi(R^2 - r^2) \cdot y \\ &= 2\pi[(4 - x + \delta x)^2 - (4 - x)^2] \sqrt{9 - x^2} \\ &= 2\pi[(4 - x)^2 + 2(4 - x)\delta x + (\delta x)^2 - (4 - x)^2] \\ &\quad \times \sqrt{9 - x^2} \end{aligned}$$

$$\text{As } (\delta x)^2 \rightarrow 0$$

$$= 2\pi[2(4 - x)\delta x] \sqrt{9 - x^2}$$

$$= 4\pi(4 - x)\sqrt{9 - x^2} \delta x$$

$$V = \lim_{\delta x \rightarrow 0} \sum_{x=-3}^3 4\pi(4 - x)\sqrt{9 - x^2} \delta x$$

$$= 4\pi \int_{-3}^3 (4 - x)\sqrt{9 - x^2} dx$$

D

# MATHEMATICS EXTENSION 2 – QUESTION MC

SUGGESTED SOLUTIONS

MARKS

MARKER'S COMMENTS

10.

$$\frac{dv}{dt} = -(1 + v)$$

$$\frac{dv}{1 + v} = -1$$

$$\int_0^t \frac{dv}{1 + v} = \int_u^0 \frac{-1}{1 + v} dv$$

$$t = -[\ln(1 + v)]_u^0$$

$$= -(0 - \ln(1 + u))$$

$$t = \ln(1 + u)$$

C

# MATHEMATICS EXTENSION 2 – QUESTION I

| SUGGESTED SOLUTIONS                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       | MARKS                                                           | MARKER'S COMMENTS                                                                                                                                                                                                    |
|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <p>a) i) <math>\arg(w-z) = \arg(2+2i-(3-i))</math></p> <p><math>= \arg(2+2i-3-i)</math></p> <p><math>= \arg(-1+i)</math></p> <p><math>= \tan^{-1}(-i)</math></p> <p><math>= \frac{3\pi}{4}</math></p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     | <p><math>\frac{1}{2}</math></p> <p><math>\frac{1}{2}</math></p> | <p>Many students gave the answer as <math>-\frac{\pi}{4}</math> not realising that <math>(-1+i)</math> is in the 2nd Quad  (1)</p> |
| <p>ii) <math>2+2i + 2i(3+i) = 2+2i+6i-2</math></p> <p><math>= 2+2i-2-6i</math></p> <p><math>= -4i</math></p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              | <p>1</p> <p>1</p>                                               | <p>A few student took the conjugate of <math>6i-2</math> as <math>6i+2</math> which is incorrect. (2)</p>                                                                                                            |
| <p>b) Use the substitution <math>x = \sin \theta</math></p> <p><math>\frac{dx}{d\theta} = \cos \theta</math></p> <p><math>dx = \cos \theta d\theta</math></p> <p>when <math>x=1, \theta = \frac{\pi}{2}</math></p> <p>when <math>x=0, \theta = 0</math></p> <p><math>\int_0^1 \sqrt{1-x^2} dx = \int_0^{\frac{\pi}{2}} \sqrt{1-\sin^2 \theta} \cdot \cos \theta d\theta</math></p> <p><math>= \int_0^{\frac{\pi}{2}} \cos^2 \theta \cos \theta d\theta</math></p> <p><math>= \int_0^{\frac{\pi}{2}} \cos^2 \theta d\theta</math></p> <p><math>= \int_0^{\frac{\pi}{2}} \frac{1}{2} + \frac{1}{2} \cos 2\theta d\theta</math></p> <p><math>= \left[ \frac{1}{2} \theta + \frac{1}{4} \sin 2\theta \right]_0^{\frac{\pi}{2}}</math></p> <p><math>= \frac{\pi}{4}</math></p> | <p>1</p> <p>1</p> <p>1</p>                                      | <p>Some errors were made with the substitution otherwise this question was well answered. (3)</p>                                                                                                                    |

# MATHEMATICS EXTENSION 2 – QUESTION II

| SUGGESTED SOLUTIONS                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               | MARKS                                                                                                                                           | MARKER'S COMMENTS                                                                         |
|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------|
| <p>c) i)</p> <p><math>\frac{x}{(x-1)(x^2+1)} = \frac{A}{x+1} + \frac{Bx+C}{x^2+1}</math></p> <p><math>x = A(x^2+1) + (Bx+C)(x+1)</math></p> <p>when <math>x = -1</math></p> <p><math>-1 = 2A</math></p> <p><math>A = -\frac{1}{2}</math></p> <p>when <math>x = 0</math></p> <p><math>0 = -\frac{1}{2}(1) + (0+C)(0+1)</math></p> <p><math>0 = -\frac{1}{2} + C</math></p> <p><math>C = \frac{1}{2}</math></p> <p>when <math>x = 1</math></p> <p><math>1 = -\frac{1}{2}(1+1) + (B+\frac{1}{2})(1+1)</math></p> <p><math>1 = -1 + 2B + 1</math></p> <p><math>2B = 1</math></p> <p><math>B = \frac{1}{2}</math></p> <p><math>\therefore \frac{x}{(x+1)(x^2+1)} = \frac{-1}{2(x+1)} + \frac{x+1}{2(x^2+1)}</math></p> | <p></p> <p></p> <p></p> <p></p> <p></p> <p><math>\frac{1}{2}</math></p> <p><math>\frac{1}{4}</math></p> <p></p> <p><math>\frac{1}{4}</math></p> | <p>well answered.</p> <p></p> <p></p> <p></p> <p></p> <p></p> <p></p> <p></p> <p>(2)</p>  |
| <p>ii) <math>\int \frac{x dx}{(x+1)(x^2+1)} = \frac{-1}{2} \int \frac{dx}{x+1} + \frac{1}{2} \int \frac{x+1}{x^2+1} dx</math></p> <p><math>= \frac{-1}{2} \int \frac{dx}{x+1} + \frac{1}{2} \int \frac{x}{x^2+1} dx + \frac{1}{2} \int \frac{1}{x^2+1} dx</math></p> <p><math>= \left[ -\frac{1}{2} \ln x+1  + \frac{1}{4} \ln x^2+1  + \frac{1}{2} \tan^{-1} x \right] + C</math></p>                                                                                                                                                                                                                                                                                                                            | <p></p> <p></p> <p></p> <p></p>                                                                                                                 | <p>For expression written with A, B and C.</p> <p>Overall quite well done.</p> <p>(2)</p> |

# MATHEMATICS EXTENSION 2 – QUESTION 11

| SUGGESTED SOLUTIONS                              | MARKS         | MARKER'S COMMENTS       |
|--------------------------------------------------|---------------|-------------------------|
| d) $(a+ib)^2 = -3+4i$                            |               |                         |
| $a^2+2iba-b^2 = -3+4i$                           |               |                         |
| Equating parts:                                  |               |                         |
| $a^2-b^2 = -3$ (1)                               |               |                         |
| $2ab = 4$ (2)                                    |               |                         |
| From (2) $a = \frac{2}{b}$ sub in (1)            | 1             |                         |
| $\frac{4}{b^2} - b^2 = -3$                       |               |                         |
| $4 - b^4 = -3b^2$                                |               |                         |
| $b^4 - 3b^2 - 4 = 0$                             |               |                         |
| $(b^2-4)(b^2+1) = 0$                             |               | 1 mark was              |
| $b^2 = 4$ or $b^2 = -1$ no solution as b is real |               | lost if students        |
| $\therefore b = \pm 2$                           | 1             | factorised incorrectly  |
| when $b=2$ , $a=1$                               |               |                         |
| when $b=-2$ , $a=-1$                             | 1             | Some students           |
| ii) $z = \frac{3 \pm \sqrt{9-4(3-i)}}{2}$        |               | did not realise         |
| $= \frac{3 \pm \sqrt{9-12+4i}}{2}$               |               | (3) that since $ab > 0$ |
| $= \frac{3 \pm \sqrt{-3+4i}}{2}$ from            |               | the $a > 0$ & $b > 0$   |
| $= \frac{3 \pm (1+2i)}{2}$ from (i)              | $\frac{1}{2}$ | (or $a < 0$ & $b < 0$ ) |
| $z = \frac{3+1+2i}{2}$ or $z = \frac{3-1-2i}{2}$ |               | $\frac{1}{2}$ mark was  |
| $= \frac{4+2i}{2}$ or $= \frac{2-2i}{2}$         |               | taken off if            |
| $\therefore z = 2+i$ or $z = 1-i$                | 1             | student did not         |
|                                                  |               | clearly show this       |

# MATHEMATICS EXTENSION 2 – QUESTION 12

| SUGGESTED SOLUTIONS                                                                         | MARKS | MARKER'S COMMENTS |
|---------------------------------------------------------------------------------------------|-------|-------------------|
| a) (i) $y = 18x^{-1}$                                                                       |       |                   |
| $\frac{dy}{dx} = -18x^{-2}$                                                                 |       |                   |
| $\therefore$ gradient of tangent at $P(2,9)$ is $-\frac{18}{2^2} = -\frac{9}{2}$            |       |                   |
| $\therefore$ gradient of normal at P is $\frac{2}{9}$                                       | 1     |                   |
| Equation of $N_1$ is                                                                        |       |                   |
| $y-9 = \frac{2}{9}(x-2)$                                                                    | 1     |                   |
| $9y-81 = 2x-4$                                                                              |       |                   |
| $2x-9y+77=0$ (1)                                                                            |       |                   |
| (ii) gradient of normal at $Q(9,2)$ is $\frac{9}{2}$                                        |       |                   |
| Equation of $N_2$ is                                                                        |       |                   |
| $y-2 = \frac{9}{2}(x-9)$                                                                    |       |                   |
| $2y-4 = 9x-81$                                                                              |       |                   |
| $9x-2y-77=0$ (2)                                                                            | 1     |                   |
| (iii) Multiply (1) by 2                                                                     |       |                   |
| $4x-18y+154=0$ (1a)                                                                         |       |                   |
| (2) $\times 9$                                                                              |       |                   |
| $81x-18y-693=0$ (2a)                                                                        |       |                   |
| (1a) - (2a)                                                                                 |       |                   |
| $77x - 847 = 0$                                                                             |       |                   |
| $x = 11$                                                                                    |       |                   |
| sub. into (1)                                                                               |       |                   |
| $22 - 9y + 77 = 0$                                                                          |       |                   |
| $9y = 99$                                                                                   |       |                   |
| $y = 11$                                                                                    |       |                   |
| $\therefore$ Coordinates of B (11, 11)                                                      | 1     |                   |
| (iv) $O(0,0)$ , $P(2,9)$ , $Q(9,2)$ and $B(11,11)$                                          |       |                   |
| $m_{OP} = \frac{9}{2}$ $m_{BP} = \frac{2}{9}$ $m_{OQ} = \frac{2}{9}$ $m_{BQ} = \frac{9}{2}$ |       |                   |



# MATHEMATICS EXTENSION 2 – QUESTION 12

## SUGGESTED SOLUTIONS

MARKS

MARKER'S COMMENTS

c) continued

$$\begin{aligned} V &= 2\pi \int_0^2 (4x - 4x^2 + x^3) dx \\ &= 2\pi \left[ 2x^2 - \frac{4}{3}x^3 + \frac{x^4}{4} \right]_0^2 \\ &= 2\pi \left( 8 - \frac{32}{3} + 4 - 0 \right) \\ &= \frac{8\pi}{3} \end{aligned}$$

1

1 mark for integration and correct calculation

d) Let  $z = x + iy$   $|z| = |x + iy|$   
 $= \sqrt{x^2 + y^2}$   
 $|z - 2| = |x + iy - 2|$   
 $= |x - 2 + iy|$   
 $= \sqrt{(x - 2)^2 + y^2}$

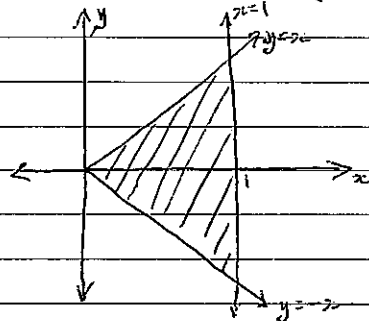
$$\begin{aligned} |z| &\leq |z - 2| \\ \sqrt{x^2 + y^2} &\leq \sqrt{(x - 2)^2 + y^2} \\ x^2 + y^2 &\leq (x - 2)^2 + y^2 \\ &\leq x^2 - 4x + 4 + y^2 \\ 0 &\leq -4x + 4 \end{aligned}$$

$$4x \leq 4$$

$$x \leq 1$$

$$\arg z \leq \frac{\pi}{4}$$

$$\arg z \geq -\frac{\pi}{4}$$



1

for region  $x \leq 1$

1

for less than  $y = x$

1

for greater than  $y = -x$

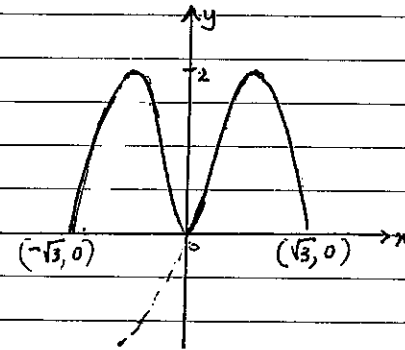
# MATHEMATICS EXTENSION 2 – QUESTION 13

## SUGGESTED SOLUTIONS

MARKS

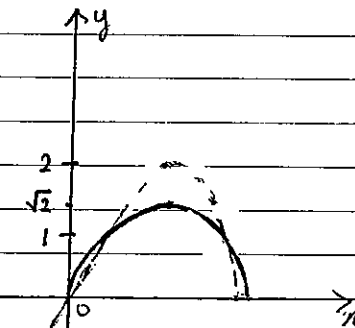
MARKER'S COMMENTS

13a) i)



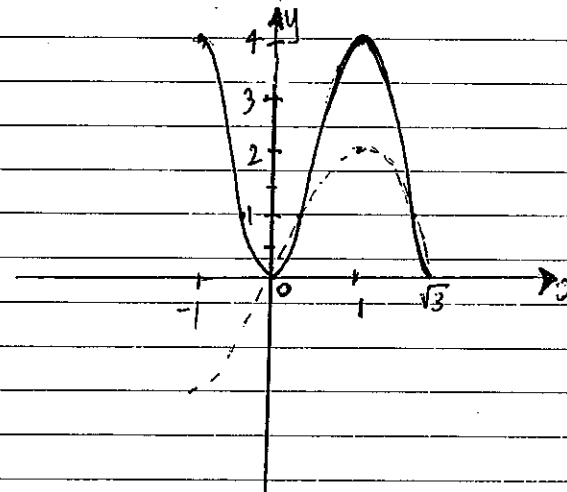
1 mark - point  $(-1, 2)$  &  $(1, 2)$   
 1 mark - shape

2



1 mark - shape  
 $\frac{1}{2}$  mark - behaviour between 0 & 1  
 (i.e.  $\sqrt{f(x)} > f(x)$ )  
 $\frac{1}{2}$  mark - point  $(1, \sqrt{2})$

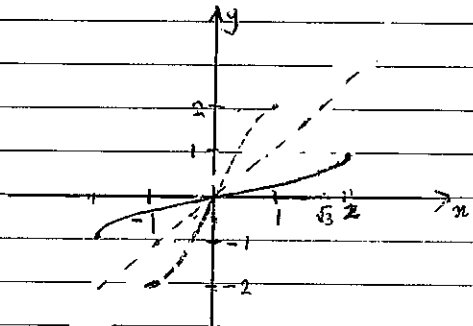
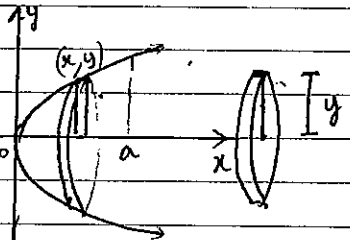
2



$\frac{1}{2}$  mark - shape  
 $\frac{1}{2}$  mark - concavity change at  $x = 0$  &  $x = \sqrt{3}$   
 1 mark for 2 points  $(1, 4)$ ,  $(-1, 4)$

2

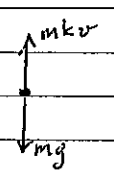
# MATHEMATICS EXTENSION 2 – QUESTION 13

| SUGGESTED SOLUTIONS                                                                                                                                                                                                                                                                                                  | MARKS                      | MARKER'S COMMENTS                                                                                                                                                   |
|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <p>iv)</p>  <p>Restricted domain: <math>-1 \leq x \leq 1</math></p> <p>Many students did not restrict the domain and sketched the inverse relation. <math>-\frac{1}{2}</math> marks.</p>                                            |                            | <p>This part was not well done.</p> <p>(2)</p>                                                                                                                      |
| <p>13b)</p>  $\delta V = \pi y^2 \delta x$ $= \pi (4ax - x^2) \delta x$ $V = \lim_{\delta x \rightarrow 0} \sum_{n=0}^a \pi (4ax - x^2) \delta x$ $= 4a\pi \int_0^a x dx$ $= 4a\pi \left[ \frac{x^2}{2} \right]_0^a$ $V = 2a^3\pi$ | <p>1</p> <p>1</p> <p>1</p> | <p>Some students took the slice parallel to the axis of rotation. Volumes by slicing requires the slice to be perpendicular to the axis of rotation.</p> <p>(3)</p> |

# MATHEMATICS EXTENSION 2 – QUESTION 13

| SUGGESTED SOLUTIONS                                                                                                                                                                                                                                                                                                                                                                                                                                                                      | MARKS | MARKER'S COMMENTS                      |
|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------|----------------------------------------|
| <p>13c)</p> $I_n = \int_1^e (1 - \log_e x)^n \cdot \frac{1}{x} dx$ $u = (1 - \log_e x)^n$ $u' = n(1 - \log_e x)^{n-1} \cdot \left(-\frac{1}{x}\right)$ $v' = 1$ $v = x$ $\therefore I_n = \left[ x(1 - \log_e x)^n \right]_1^e - \int_1^e x \cdot n(1 - \log_e x)^{n-1} \cdot \left(-\frac{1}{x}\right) dx$ $= \left[ e(1 - \log_e e)^n - 1(1 - \log_e 1)^n \right] + \int_1^e n(1 - \log_e x)^{n-1} dx$ $= -1 + \int_1^e n(1 - \log_e x)^{n-1} dx$ $= -1 + n I_{n-1}, \quad n=1, 2, 3.$ |       | <p>1 mark</p> <p>1 mark</p> <p>(2)</p> |
| <p>ii)</p> $I_3 = -1 + 3 I_2$ $= -1 + 3(-1 + 2 I_1)$ $= -1 - 3 + 6 I_1$ $= -4 + 6 I_1$ $= -4 + 6(-1 + I_0)$ $= -4 - 6 + 6 I_0$ $= -10 + 6 I_0$ $\text{but } I_0 = \int_1^e \frac{1}{x} dx$ $= x \Big _1^e$ $= e - 1$ $\therefore I_3 = -10 + 6(e - 1)$ $= -10 + 6e - 6$ $= -16 + 6e$ $\therefore \int_1^e (1 - \log_e x)^3 = -16 + 6e$                                                                                                                                                   |       | <p>1 mark</p> <p>1 mark</p> <p>(2)</p> |

# MATHEMATICS EXTENSION 2 – QUESTION 14

| SUGGESTED SOLUTIONS                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            | MARKS | MARKER'S COMMENTS                                                    |
|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------|----------------------------------------------------------------------|
| <p>a)(i) Forces acting on the particle</p>  <p>Net force <math>m\ddot{x} = mg - mkv</math></p> <p><math>\ddot{x} = g - kv</math></p> <p>As <math>\ddot{x} \rightarrow 0</math> <math>v \rightarrow \frac{g}{k}</math></p> <p>For terminal velocity <math>V</math> <math>\ddot{x} = 0</math>, <math>v = V</math></p> <p><math>\therefore V = \frac{g}{k}</math></p> <p><math>k = \frac{g}{V}</math></p> <p><math>\therefore \ddot{x} = g - \frac{g}{V}v</math></p> <p><math>= \frac{gV - gv}{V}</math></p> <p><math>= \frac{g}{V}(V - v)</math></p>                                                                                                                                                                            |       |                                                                      |
| <p>(ii) Initial conditions <math>t=0</math>, <math>x=0</math>, <math>v=0</math></p> <p>when <math>t=T</math>, <math>x=X</math>, <math>v=\frac{V}{2}</math></p> <p><math>\ddot{x} = \frac{g}{V}(V - v)</math></p> <p><math>v \frac{dv}{dx} = \frac{g}{V}(V - v)</math></p> <p><math>\frac{dv}{dx} = \frac{g(V - v)}{Vv}</math></p> <p><math>\frac{dx}{dv} = \frac{Vv}{g(V - v)}</math></p> <p><math>\int_0^X dx = \int_0^{\frac{V}{2}} \frac{Vv}{g(V - v)} dv</math></p> <p><math>X = \frac{V}{g} \int_0^{\frac{V}{2}} \frac{v}{V - v} dv</math></p> <p><math>= -\frac{V}{g} \int_0^{\frac{V}{2}} \frac{-v}{V - v} dv</math></p> <p><math>= -\frac{V}{g} \int_0^{\frac{V}{2}} \frac{V - v - V}{V - v} dv</math></p> <p><math>= -\frac{V}{g} \int_0^{\frac{V}{2}} \left(1 - \frac{V}{V - v}\right) dv</math></p> | 1     | 1 mark for stating that terminal velocity occurs when $\ddot{x} = 0$ |
|                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                | 1     | 1 mark for finding and using $k = \frac{g}{V}$                       |
|                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                | 1     | 1 mark for setting up this integral                                  |

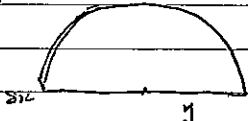
# MATHEMATICS EXTENSION 2 – QUESTION 14

| SUGGESTED SOLUTIONS                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        | MARKS | MARKER'S COMMENTS                               |
|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------|-------------------------------------------------|
| <p>a)(ii) continued</p> <p><math>X = -\frac{V}{g} \left[ \left[ v \right]_0^{\frac{V}{2}} + V \int_0^{\frac{V}{2}} \frac{-dv}{V - v} \right]</math></p> <p><math>= -\frac{V}{g} \left( \frac{V}{2} + V \left[ \ln(V - v) \right]_0^{\frac{V}{2}} \right)</math></p> <p><math>= -\frac{V^2}{2g} - \frac{V^2}{g} (\ln(V - \frac{V}{2}) - \ln V)</math></p> <p><math>= -\frac{V^2}{2g} - \frac{V^2}{g} \ln \frac{V}{2V}</math></p> <p><math>= -\frac{V^2}{2g} - \frac{V^2}{g} \ln \frac{1}{2}</math></p> <p><math>= -\frac{V^2}{2g} + \frac{V^2}{g} \ln 2</math> ①</p> <p><math>\ddot{x} = \frac{g}{V}(V - v)</math></p> <p><math>\frac{dv}{dt} = \frac{g}{V}(V - v)</math></p> <p><math>\frac{dt}{dv} = \frac{V}{g(V - v)}</math></p> <p><math>\int_0^T dt = \int_0^{\frac{V}{2}} \frac{V}{g(V - v)} dv</math></p> <p><math>T = -\frac{V}{g} \int_0^{\frac{V}{2}} \frac{-dv}{V - v}</math></p> <p><math>= -\frac{V}{g} \left[ \ln(V - v) \right]_0^{\frac{V}{2}}</math></p> <p><math>= -\frac{V}{g} (\ln(V - \frac{V}{2}) - \ln V)</math></p> <p><math>= -\frac{V}{g} \ln \frac{V}{2V}</math></p> <p><math>= -\frac{V}{g} \ln \frac{1}{2}</math></p> <p><math>= \frac{V}{g} \ln 2</math></p> | 1     | 1 mark for finding this integral                |
|                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            | 1     | 1 mark for setting up and finding this integral |

**MATHEMATICS EXTENSION 2 – QUESTION 14**

| MATHEMATICS EXTENSION 2 – QUESTION 14                                                                                                                                                                                                                                                                                                                                                                         |       |                                                                |
|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------|----------------------------------------------------------------|
| SUGGESTED SOLUTIONS                                                                                                                                                                                                                                                                                                                                                                                           | MARKS | MARKER'S COMMENTS                                              |
| <p>a) (ii) continued</p> $VT = \frac{v^2 \ln 2}{g}$ <p>from ①</p> $VT - X = \frac{v^2 \ln 2}{g} + \frac{v^2}{2g} - \frac{v^2 \ln 2}{g}$ $VT - X = \frac{v^2}{2g}$                                                                                                                                                                                                                                             | 1     | 1 mark for combining the two integrals                         |
| <p>b)</p>                                                                                                                                                                                                                                                                                                                                                                                                     | 3     | 1 mark for each arm and 1 mark for vertical asymptote at $x=0$ |
| <p>c)</p> $2xy + x^2 \frac{dy}{dx} + y^2 + 2y \frac{dy}{dx} = 0$ $x^2 \frac{dy}{dx} + 2y \frac{dy}{dx} = -2xy - y^2$ $(x^2 + 2y) \frac{dy}{dx} = -(2xy + y^2)$ $\frac{dy}{dx} = -\frac{(2xy + y^2)}{x^2 + 2y}$ <p>at <math>(1, 2)</math> <math>\frac{dy}{dx} = -\frac{(2 \times 1 \times 2 + 2^2)}{1^2 + 2 \times 2}</math></p> $= -\frac{8}{5}$ <p>Equation of the tangent</p> $y - 2 = -\frac{8}{5}(x - 1)$ | 1     | 1 mark for implicit differentiation                            |
|                                                                                                                                                                                                                                                                                                                                                                                                               | 1     | 1 mark for gradient of tangent                                 |

**MATHEMATICS EXTENSION 2 – QUESTION 14**

| SUGGESTED SOLUTIONS                                                                                                                                                                                      |   | MARKS                                          | MARKER'S COMMENTS |
|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---|------------------------------------------------|-------------------|
| c) continued                                                                                                                                                                                             |   |                                                |                   |
| $5y - 10 = -8x + 8$                                                                                                                                                                                      |   |                                                |                   |
| $8x + 5y - 18 = 0$                                                                                                                                                                                       | 1 | 1 mark for equation of tangent                 |                   |
| d)                                                                                                                                                                                                       |   |                                                |                   |
|  $\delta V = \frac{1}{2} \pi r^2 \delta x$ $= \frac{1}{2} \pi y^2 \delta x$ $= \frac{1}{2} \pi (9x - x^3)^2 \delta x$ | 1 | 1 mark for volume of slice                     |                   |
| $V = \lim_{\delta x \rightarrow 0} \sum_{x=0}^3 \frac{1}{2} \pi (9x - x^3)^2 \delta x$                                                                                                                   |   |                                                |                   |
| $= \int_0^3 \frac{1}{2} \pi (9x - x^3)^2 dx$                                                                                                                                                             |   |                                                |                   |
| $= \frac{\pi}{2} \int_0^3 (81x^2 - 18x^4 + x^6) dx$                                                                                                                                                      | 1 | 1 mark for setting up the integral             |                   |
| $= \frac{\pi}{2} \left[ 27x^3 - \frac{18}{5}x^5 + \frac{x^7}{7} \right]_0^3$                                                                                                                             |   |                                                |                   |
| $= \frac{\pi}{2} \left( 27^2 - \frac{18}{5}(3^5) + \frac{(3)^7}{7} - 0 \right)$                                                                                                                          |   |                                                |                   |
| $= \frac{2916\pi}{35}$                                                                                                                                                                                   | 1 | 1 mark for integrating and finding the volume. |                   |

# MATHEMATICS EXTENSION 2 – QUESTION 15

| SUGGESTED SOLUTIONS                                                                               | MARKS         | MARKER'S COMMENTS               |
|---------------------------------------------------------------------------------------------------|---------------|---------------------------------|
| 15 a) i)                                                                                          |               |                                 |
| $x = a \cos \theta$ $y = b \sin \theta$                                                           |               |                                 |
| $\frac{dx}{d\theta} = -a \sin \theta$ $\frac{dy}{d\theta} = b \cos \theta$                        |               |                                 |
| $\therefore \frac{dy}{dx} = \frac{dy}{d\theta} \times \frac{d\theta}{dx}$                         |               |                                 |
| $= \frac{b \cos \theta \times -1}{a \sin \theta}$                                                 |               |                                 |
| $= -\frac{b \cos \theta}{a \sin \theta}$                                                          |               | 1 mark for gradient             |
| Equation of tangent at P                                                                          |               |                                 |
| $y - b \sin \theta = \frac{-b \cos \theta}{a \sin \theta} (x - a \cos \theta)$                    |               | 1 mark for Equation of tangent. |
| $y a \sin \theta - a b \sin^2 \theta = -b \cos \theta x + a b \cos^2 \theta$                      |               |                                 |
| $x b \cos \theta + y a \sin \theta = a b \sin^2 \theta + a b \cos^2 \theta$                       |               |                                 |
| $= a b (\sin^2 \theta + \cos^2 \theta)$                                                           |               |                                 |
| $\therefore x b \cos \theta + y a \sin \theta = a b$                                              |               |                                 |
| $\div ab$                                                                                         |               |                                 |
| $\frac{x \cos \theta}{a} + \frac{y \sin \theta}{b} = 1$                                           |               | (2)                             |
| ii) Gradient of normal is $\frac{a \sin \theta}{b \cos \theta}$                                   | $\frac{1}{2}$ |                                 |
| Equation of normal is:                                                                            |               |                                 |
| $y - b \sin \theta = \frac{a \sin \theta}{b \cos \theta} (x - a \cos \theta)$                     | $\frac{1}{2}$ |                                 |
| $b y \cos \theta - b^2 \sin \theta \cos \theta = a x \sin \theta - a^2 \sin^2 \theta \cos \theta$ | $\frac{1}{2}$ |                                 |
| $a x \sin \theta - b y \cos \theta = a^2 \sin \theta \cos \theta - b^2 \sin \theta \cos \theta$   |               |                                 |
| $= \sin \theta \cos \theta (a^2 - b^2)$                                                           | $\frac{1}{2}$ |                                 |
| $\therefore \frac{ax}{\cos \theta} - \frac{by}{\sin \theta} = a^2 - b^2$                          |               | (2)                             |

# MATHEMATICS EXTENSION 2 – QUESTION 15

| SUGGESTED SOLUTIONS                                                         | MARKS         | MARKER'S COMMENTS          |
|-----------------------------------------------------------------------------|---------------|----------------------------|
| a) iii) To find the coordinates of A,<br>sub $x=0$ into equation of tangent |               |                            |
| $\frac{y \sin \theta}{b} = 1$                                               |               |                            |
| $y = \frac{b}{\sin \theta}$                                                 |               |                            |
| $\therefore A$ is $(0, \frac{b}{\sin \theta})$                              | 1             |                            |
| To find the coordinates of B<br>sub $x=0$ into equation of normal           |               |                            |
| $\frac{-by}{\sin \theta} = a^2 - b^2$                                       |               |                            |
| $\frac{by}{\sin \theta} = b^2 - a^2$                                        |               |                            |
| $y = \frac{(b^2 - a^2) \sin \theta}{b}$                                     |               |                            |
| $\therefore B$ is $(0, \frac{(b^2 - a^2) \sin \theta}{b})$                  | 1             |                            |
| Gradient of AS: $m_{AS} = \frac{b}{\sin \theta} - 0$                        |               |                            |
| $= \frac{a - ae}{a \sin \theta}$                                            |               |                            |
| $= \frac{-b}{a \sin \theta}$                                                |               |                            |
| Gradient of BS: $m_{BS} = \frac{(b^2 - a^2) \sin \theta}{-aeb}$             |               |                            |
| $= \frac{(a^2 - b^2) \sin \theta}{aeb}$                                     |               |                            |
| From $b^2 = a^2(1 - e^4)$<br>$= a^2 - a^2 e^4$                              |               |                            |
| $\therefore m_{BS} = \frac{a^2 e^2 \sin \theta}{aeb}$                       |               | s. $a^2 - b^2 = a^2 e^2$ . |
| $= \frac{ae \sin \theta}{b}$                                                | $\frac{1}{2}$ |                            |

# MATHEMATICS EXTENSION 2 – QUESTION 15

| SUGGESTED SOLUTIONS                                                                                                                                                                                     | MARKS | MARKER'S COMMENTS                                                                                                                                                   |
|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 15a) iii)                                                                                                                                                                                               |       |                                                                                                                                                                     |
| Now                                                                                                                                                                                                     |       |                                                                                                                                                                     |
| $m_{AS} \times m_{BS} = \frac{-b}{a \sin \theta} \times \frac{a \sin \theta}{b}$                                                                                                                        | 1/2   |                                                                                                                                                                     |
| $= -1$                                                                                                                                                                                                  |       |                                                                                                                                                                     |
| $\therefore AS$ is perpendicular to $BS$ .                                                                                                                                                              |       | (3)                                                                                                                                                                 |
| iv) $\angle APB = 90^\circ$ (tangents and normals at right angles to each other)<br>$\angle ASB = 90^\circ$ (proven above)<br>$\therefore \angle APB = \angle ASB$                                      |       |                                                                                                                                                                     |
| Since the interval $AB$ subtends equal angles at two points, at $P$ and $S$ , on the same side of it,<br>$\therefore A, B$ and $S, P$ form a cyclic quadrilateral<br>$\therefore$ are concyclic points. | 1     | Many students gave the reason the angles in the same segment are equal.<br>This reason applies only if the points are already concyclic.<br>Care needs to be taken. |
|                                                                                                                                                                                                         |       | (1)                                                                                                                                                                 |

# MATHEMATICS EXTENSION 2 – QUESTION 15

| SUGGESTED SOLUTIONS                                                                           | MARKS | MARKER'S COMMENTS                                                                   |
|-----------------------------------------------------------------------------------------------|-------|-------------------------------------------------------------------------------------|
| 15(b)                                                                                         |       |                                                                                     |
| i) $\frac{d(\frac{1}{2}v^2)}{dx} = \frac{1}{2} \left(1 - \frac{1}{x^2}\right)$                |       |                                                                                     |
| $\frac{1}{2}v^2 = \frac{1}{2} \int \left(1 - \frac{1}{x^2}\right) dx$                         |       |                                                                                     |
| $= \frac{1}{2} \left[ x + \frac{1}{x} \right] + C$                                            |       |                                                                                     |
| $v^2 = x + \frac{1}{x} + C$                                                                   | 1     |                                                                                     |
| when $x=1, v=2$<br>$4 = 1 + 1 + C$<br>$C = 2$                                                 |       |                                                                                     |
| $\therefore v^2 = x + \frac{1}{x} + 2$<br>$= \frac{x^2 + 2x + 1}{x}$<br>$= \frac{(x+1)^2}{x}$ | 1     |                                                                                     |
| $v = \pm \frac{x+1}{\sqrt{x}} \Rightarrow \frac{dx}{dt} = \pm \frac{1+x}{\sqrt{x}}$           |       |                                                                                     |
| but when $x=1, v=2 > 0$<br>$\therefore \frac{dx}{dt} = \frac{1+x}{\sqrt{x}}$                  |       |                                                                                     |
|                                                                                               | 1     | Many students did not state that $v > 0$ since $x=1$<br>$\therefore$ lost 1/2 marks |
|                                                                                               |       | (3)                                                                                 |

MATHEMATICS EXTENSION 2 – QUESTION 15

| SUGGESTED SOLUTIONS                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             | MARKS                                                                    | MARKER'S COMMENTS |
|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------|-------------------|
| $b) \text{ ii) } \frac{dt}{dx} = \frac{\sqrt{x}}{1+x}$ $\int_0^t dt = \int_1^{\sqrt{3}} \frac{\sqrt{x}}{1+x} dx$ $t = \int_1^{\sqrt{3}} \frac{\sqrt{x}}{1+x} dx$ $\text{Let } u = \sqrt{x}, \text{ i.e. } x = u^2$ $\frac{dx}{du} = 2u$ $\text{when } x = 3, u = \sqrt{3}$ $x = 1, u = 1$ $\therefore t = \int_1^{\sqrt{3}} \frac{u \cdot 2u du}{1+u^2}$ $= \int_1^{\sqrt{3}} \frac{2u^2}{1+u^2} du$ $= \int_1^{\sqrt{3}} \frac{2u^2 + 2 - 2}{1+u^2} du$ $= \int_1^{\sqrt{3}} \left( 2 - \frac{2}{1+u^2} \right) du$ $= [2u - 2 \tan^{-1} u]_1^{\sqrt{3}}$ $= 2[(\sqrt{3} - 2 \tan^{-1} \sqrt{3}) - (1 - \tan^{-1} 1)]$ $= 2[\sqrt{3} - 2 \tan^{-1} \sqrt{3} - 1 + \frac{\pi}{4}]$ $= 2\sqrt{3} - 4 \tan^{-1} \sqrt{3} - 2 + \frac{\pi}{2}$ $= 2\sqrt{3} - 4 \cdot \frac{\pi}{6} - 2 + \frac{\pi}{2}$ $= 2\sqrt{3} - 2 - \frac{\pi}{6}$ $\div 0.94 \text{ sec}$ | <p><math>\frac{1}{2}</math></p> <p><math>\frac{1}{2}</math></p> <p>1</p> | <p>(4)</p>        |

MATHEMATICS EXTENSION 2 – QUESTION 16

| SUGGESTED SOLUTIONS                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 | MARKS                                                                   | MARKER'S COMMENTS                                                                                                                                                                                                                                  |
|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| $a) \text{ (i) } \vec{OR} = \frac{1}{2} \vec{OP} \quad \text{①}$ $\vec{QP}$ is a $90^\circ$ clockwise rotation of $\vec{OQ}$<br>multiplied by a factor of $k, k > 0$<br>$\vec{QP} = -i \vec{OQ} \times k$ $= -ki \vec{OQ}$ $= -kiw$ $\vec{OP} = \vec{OQ} + \vec{QP}$ $= w + (-kiw)$ $\text{From ① } \vec{OR} = \frac{1}{2} \vec{OP}$ $= \frac{1}{2}(w - kiw)$ $= \frac{1}{2}(1 - ki)w$ $\text{(ii) } \vec{RQ} = \vec{OQ} - \vec{OR}$ $= w - \frac{1}{2}(1 - ki)w$ $= w - \frac{w}{2} + \frac{kiw}{2}$ $= \frac{w}{2} + \frac{kiw}{2}$ $= \frac{w}{2}(1 + ki)$ $ \vec{OR}  = \left  \frac{1}{2}(1 - ki)w \right $ $= \left  \frac{1}{2} \right   1 - ki   w $ $= \frac{1}{2} \sqrt{1 + k^2}  w $ $= \frac{1}{2} \sqrt{1 + k^2}  w $ $ \vec{RQ}  = \left  \frac{1}{2}(1 + ki)w \right $ $= \left  \frac{1}{2} \right   1 + ki   w $ $= \frac{1}{2} \sqrt{1 + k^2}  w $ $=  \vec{OR} $ | <p>1</p> <p>1</p> <p>1</p> <p>1</p> <p>1</p> <p>1</p> <p>1</p> <p>1</p> | <p>1 mark for finding <math>\vec{QP}</math> in terms of <math>w</math> and <math>k</math></p> <p>1 mark for finding <math>\vec{OP}</math> in terms of <math>w</math></p> <p>1 mark for relating <math>\vec{OR}</math> to <math>\vec{OP}</math></p> |

MATHEMATICS EXTENSION 2 – QUESTION 16

| SUGGESTED SOLUTIONS                                                                                     | MARKS | MARKER'S COMMENTS                           |
|---------------------------------------------------------------------------------------------------------|-------|---------------------------------------------|
| b) Let $\alpha = \tan^{-1}\left(\frac{x+1}{x-1}\right)$ and $\beta = \tan^{-1}\left(\frac{1}{x}\right)$ |       |                                             |
| $\tan \alpha = \frac{x+1}{x-1}$ $\tan \beta = \frac{1}{x}$ $x > 1$                                      | 1     | 1 mark for substitution                     |
| $\tan(\alpha - \beta) = \frac{\tan \alpha - \tan \beta}{1 + \tan \alpha \tan \beta}$                    |       |                                             |
| $= \frac{\frac{x+1}{x-1} - \frac{1}{x}}{1 + \left(\frac{x+1}{x-1}\right)\left(\frac{1}{x}\right)}$      | 1     | 1 mark for finishing                        |
| $= \frac{x(x+1) - (x-1)}{x(x-1)} = \frac{x(x+1) + x(x-1)}{x(x-1)}$                                      |       | 1 mark for finishing                        |
| $= \frac{x^2+1}{x(x-1)} \times \frac{x(x-1)}{x^2+1}$                                                    |       |                                             |
| $= 1$                                                                                                   |       |                                             |
| $\alpha - \beta = \tan^{-1}(1)$                                                                         |       |                                             |
| $= \frac{\pi}{4}$                                                                                       | 1     | 1 mark for $\alpha - \beta = \frac{\pi}{4}$ |
| $\therefore \tan^{-1} \frac{x+1}{x-1} - \tan^{-1} \frac{1}{x} = \frac{\pi}{4}$                          |       |                                             |
| c) (i) when $t=0$ , $x=0$ , $v=v_0$                                                                     |       |                                             |
| $m\ddot{x} = -kv$                                                                                       |       |                                             |
| $\ddot{x} = -\frac{kv}{m}$                                                                              |       |                                             |
| $\ddot{x} = \frac{dv}{dt}$                                                                              |       |                                             |
| $\frac{dv}{dt} = -\frac{kv}{m}$                                                                         |       |                                             |
| $\frac{dt}{dv} = -\frac{m}{kv}$                                                                         |       |                                             |
| $\int_0^t dt = \int_{v_0}^v -\frac{m}{kv} dv$                                                           | 1     | 1 mark for setting up the integral          |
| $t = -\frac{m}{k} \int_{v_0}^v \frac{1}{v} dv$                                                          |       |                                             |
| $= -\frac{m}{k} \left[ \ln x  \right]_{v_0}^v$                                                          |       |                                             |
| $= -\frac{m}{k} (\ln v - \ln v_0)$                                                                      | 1     | 1 mark for finding the integral             |

MATHEMATICS EXTENSION 2 – QUESTION

| SUGGESTED SOLUTIONS                                                     | MARKS | MARKER'S COMMENTS                  |
|-------------------------------------------------------------------------|-------|------------------------------------|
| e) (i) continued                                                        |       |                                    |
| $-\frac{kt}{m} = \ln \frac{v}{v_0}$                                     |       |                                    |
| $e^{-\frac{kt}{m}} = \frac{v}{v_0}$                                     | 1     | 1 mark for taking each side to e   |
| $v = v_0 e^{-\frac{kt}{m}}$                                             |       |                                    |
| (ii) $\frac{dx}{dt} = v_0 e^{-\frac{kt}{m}}$                            |       |                                    |
| $\int_0^x dx = \int_0^t v_0 e^{-\frac{kt}{m}} dt$                       | 1     | 1 mark for setting up the integral |
| $x = v_0 \int_0^t e^{-\frac{kt}{m}} dt$                                 |       |                                    |
| $= v_0 \left[ -\frac{m}{k} e^{-\frac{kt}{m}} \right]_0^t$               | 1     | 1 mark for finding the integral    |
| $= v_0 \left( -\frac{m}{k} e^{-\frac{kt}{m}} + \frac{m}{k} e^0 \right)$ |       |                                    |
| $= \frac{mv_0}{k} (-e^{-\frac{kt}{m}} + 1)$                             | 1     | 1 mark for simplification          |
| $= \frac{mv_0}{k} (1 - e^{-\frac{kt}{m}})$                              |       |                                    |
| (iii) as $t \rightarrow \infty$ , $e^{-\frac{kt}{m}} \rightarrow 0$     | 1     |                                    |
| $\therefore x \rightarrow \frac{mv_0}{k}$                               |       |                                    |
| $\therefore x_L = \frac{mv_0}{k}$                                       |       |                                    |